Quantification: 543_3
=====================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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| measure | value |
|---|---|
| lacuna count | 11 |
| interior lacuna count | 10 |
| frame-edge lacuna count | 1 |

Interior lacunae only: mean, median and sample SD (n = 10)
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| measure | kind | mean | median | sd | units |
|---|---|---|---|---|---|
| area_px2 | lacuna shape | 2350.2 | 2188.5 | 668.925 | px^2 |
| major_axis_length_px | lacuna shape | 91.5632 | 84.5141 | 33.0603 | px |
| minor_axis_length_px | lacuna shape | 36.5948 | 35.0553 | 10.2044 | px |
| aspect_ratio | lacuna shape | 2.6601 | 2.3867 | 1.276 | unitless |
| eccentricity | lacuna shape | 0.889 | 0.9075 | 0.078 | unitless |
| solidity | lacuna shape | 0.8742 | 0.9204 | 0.1013 | unitless |
| roots_count | headline | 7.7 | 6 | 4.7854 | count |
| ring_length_r30_px | headline | 356.6 | 393 | 134.17 | px |
| ring_length_r60_px | reported | 978.5 | 1040 | 310.334 | px |
| owned_length_px | ownership-dependent | 3642.1 | 4215.41 | 2042.54 | px |
| edge_count | ownership-dependent | 166.4 | 196 | 96.7106 | count |
| mean_edge_length_px | ownership-dependent | 23.1874 | 21.9775 | 3.5749 | px |
| perimeter_px | normalised | 252.45 | 233.714 | 76.1582 | px |
| ring_area_r30_px2 | normalised | 9498.1 | 8903.5 | 1857.93 | px^2 |
| ring_area_r60_px2 | normalised | 23342.5 | 22921 | 2919.57 | px^2 |
| in_frame_fraction_r30 | normalised | 0.9969 | 1 | 0.0099 | unitless |
| in_frame_fraction_r60 | normalised | 0.9761 | 1 | 0.0633 | unitless |
| ring_density_r30 | normalised | 0.0371907 | 0.041188 | 0.0118929 | px^-1 |
| ring_density_r60 | normalised | 0.0417628 | 0.0468244 | 0.0122817 | px^-1 |
| roots_per_100px_perimeter | normalised | 2.9128 | 2.7564 | 1.2447 | per 100 px |
| ring_attached_length_r30_px | network v2 | 323.7 | 364.5 | 135.904 | px |
| ring_attached_length_r60_px | network v2 | 829.2 | 943 | 349.998 | px |
| ring_length_w_r30_px | network v2 | 403.733 | 438.513 | 155.504 | px |
| ring_length_w_r60_px | network v2 | 1113.19 | 1198.64 | 355.221 | px |
| sholl_crossings_r10 | network v2 | 9.8 | 10 | 4.4422 | count |
| sholl_crossings_r20 | network v2 | 12 | 13 | 3.6515 | count |
| sholl_crossings_r30 | network v2 | 14.1 | 15 | 4.7246 | count |
| loop_count | loops, report only | 19.1 | 20.5 | 12.9825 | count |
| loop_edge_share | loops, report only | 0.3761 | 0.4212 | 0.1708 | share |
| ring_width_mean_r30_px | width | 7.6134 | 7.7868 | 0.5923 | px |

Per field
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| measure | value | units |
|---|---|---|
| analysed_area_px2 | 1.02454e+06 | px^2 |
| total_skeleton_length_px | 38294 | px |
| canalicular_length_density_per_px | 0.0373767 | px^-1 |
| junction_count | 972 | count |
| junction_density_per_px2 | 0.00094872 | px^-2 |
| lacuna_count | 11 | count |

| | | |
|---|---|---|
| lacunae_per_px2 | 1.074e-05 | px^-2 |
| skeleton_component_count | 155 | count |
| mean_component_length_px | 247.058 | px |
| median_component_length_px | 19 | px |
| field_length_density_w_per_px | 0.0425671 | px^-1 |
| field_density_without_flagged_per_px | 0.0373767 | px^-1 |
| field_density_in_roi_per_px | | px^-1 |
| unattached_length_px | 224 | px |
| unattached_component_count | 15 | count |
| canalicular_length_density_attached_per_px | 0.0373767 | px^-1 |
| field_length_density_with_unattached_per_px | 0.0375953 | px^-1 |
| loop_count | 215 | count |
| loop_edge_share | 0.4274 | share |
| edge_count | 1970 | count |

Canalicular width, the new measure

| measure | value | units |
|---|---|---|
| width_mean_px | 7.8388 | px |
| width_median_px | 8 | px |
| width_p10_px | 4.4721 | px |
| width_p90_px | 10.198 | px |
| width_pixels_used | 38294 | count |

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

| lacuna_id | area_px2 | major_axis_length_px | minor_axis_length_px | aspect_ratio | eccentricity | solidity | orientation_rad | centroid_row_px | centroid_col_px | on_border |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 2174 | 80.8669 | 35.2412 | 2.2947 | 0.9 | 0.9518 | 0.4109 | 79.3965 | 270.898 | False |
| 2 | 3447 | 90.3677 | 63.9486 | 1.4131 | 0.7066 | 0.674 | 0.4223 | 197.059 | 916.902 | False |
| 3 | 1883 | 69.2617 | 34.8694 | 1.9863 | 0.864 | 0.9448 | 0.5742 | 250.833 | 340.47 | False |
| 4 | 3536 | 181.867 | 30.1414 | 6.0338 | 0.9862 | 0.7148 | 0.0738 | 327.046 | 42.4075 | False |
| 5 | 2052 | 69.2015 | 39.4228 | 1.7554 | 0.8219 | 0.9112 | 0.0622 | 388.671 | 369.939 | False |
| 6 | 2266 | 88.1613 | 35.5691 | 2.4786 | 0.915 | 0.9021 | 0.1324 | 473.05 | 745.191 | False |
| 7 | 1625 | 80.1248 | 27.6985 | 2.8927 | 0.9383 | 0.8325 | 0.2321 | 545.572 | 877.076 | False |
| 8 | 1683 | 71.0123 | 31.3735 | 2.2634 | 0.8971 | 0.9355 | -0.0518 | 634.68 | 198.315 | False |
| 9 | 2203 | 91.2092 | 31.4053 | 2.9043 | 0.9389 | 0.9295 | 0.0832 | 734.301 | 635.44 | False |
| 10 | 2633 | 93.5592 | 36.2778 | 2.579 | 0.9218 | 0.9454 | 0.0457 | 780.576 | 481.495 | False |
| 11 | 531 | 28.7881 | 24.8252 | 1.1596 | 0.5063 | 0.9332 | 0.5942 | 1012.28 | 184.834 | True |

Network per lacuna

| lacuna_id | roots_count | ring_length_r30_px | ring_length_r60_px | owned_length_px | edge_count | mean_edge_length_px | ring_width_mean_r30_px |
|---|---|---|---|---|---|---|---|
| 1 | 7 | 336 | 1050 | 1380.05 | 63 | 21.9056 | 7.4363 |
| 2 | 15 | 572 | 1426 | 4508.26 | 221 | 20.3994 | 7.2258 |
| 3 | 8 | 380 | 1030 | 6085.64 | 276 | 22.0494 | 8.3846 |
| 4 | 16 | 458 | 976 | 5652.9 | 263 | 21.4939 | 8.1099 |
| 5 | 3 | 156 | 433 | 564.63 | 20 | 28.2315 | 6.5172 |
| 6 | 11 | 406 | 1127 | 3925.78 | 171 | 22.9578 | 8.1001 |
| 7 | 5 | 426 | 1281 | 4505.04 | 226 | 19.9338 | 6.8746 |
| 8 | 3 | 134 | 527 | 2881.56 | 139 | 20.7306 | 7.9116 |

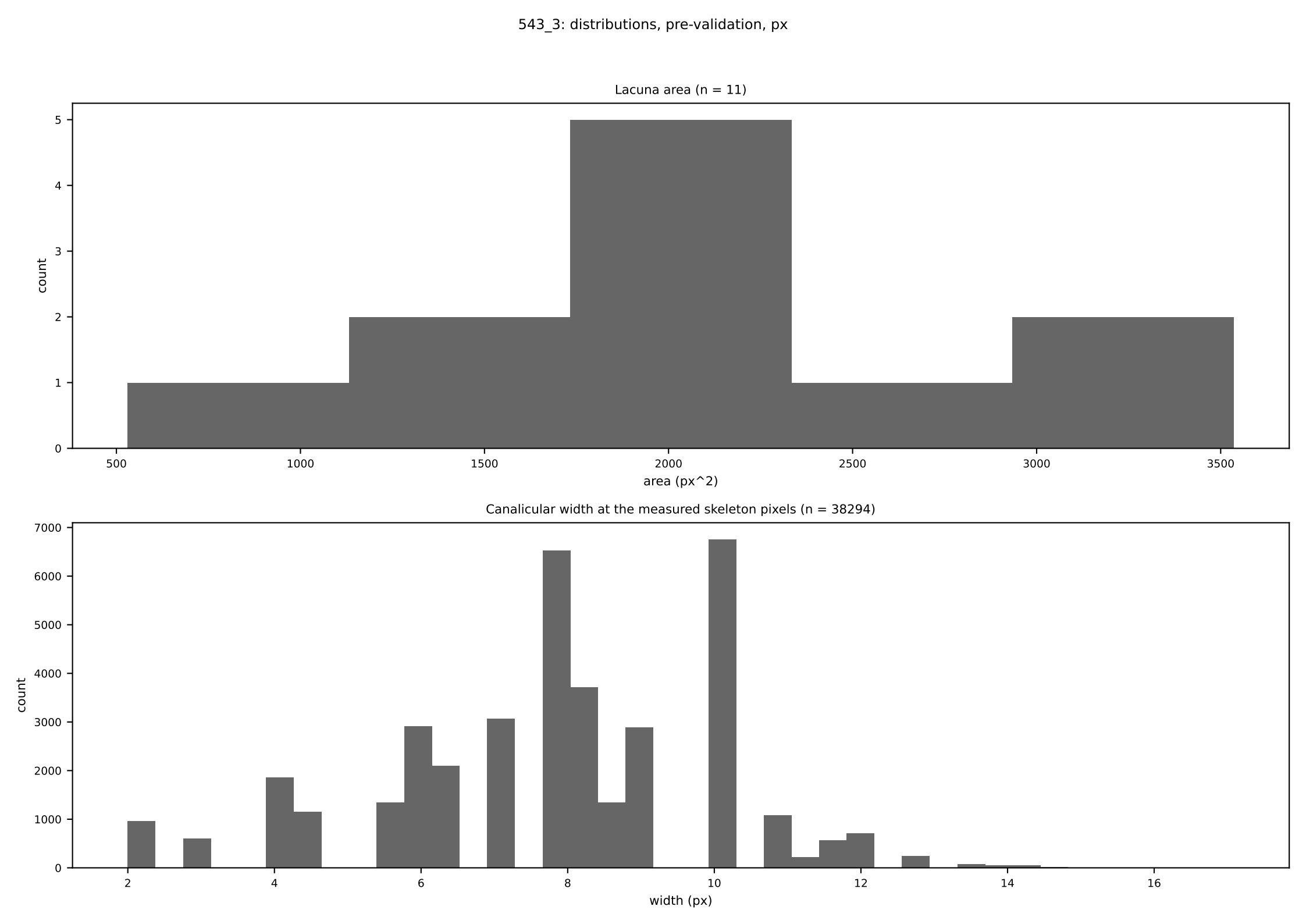
| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 9 | 5 | 409 | 1115 | 5774.27 | 248 | 23.2834 | 7.7657 |
| 10 | 4 | 289 | 820 | 1142.89 | 37 | 30.889 | 7.8079 |
| 11 | 2 | 136 | 334 | 872.009 | 42 | 20.7621 | 6.1456 |

Normalised per lacuna

| lacuna_id | perimeter_px | ring_area_r30_px2 | ring_area_r60_px2 | in_frame_fraction_r30 | in_frame_fraction_r60 | ring_density_r30 | ring_density_r60 | roots_per_100px_perimeter |
|---|---|---|---|---|---|---|---|---|
| 1 | 210.066 | 8595 | 21715 | 1 | 0.9616 | 0.0390925 | 0.0483537 | 3.3323 |
| 2 | 363.446 | 12097 | 29071 | 1 | 1 | 0.0472844 | 0.0490523 | 4.1272 |
| 3 | 184.794 | 7941 | 19515 | 1 | 1 | 0.0478529 | 0.0527799 | 4.3291 |
| 4 | 415.99 | 13536 | 26753 | 0.9687 | 0.7992 | 0.0338357 | 0.0364819 | 3.8462 |
| 5 | 205.965 | 8333 | 20507 | 1 | 1 | 0.0187208 | 0.0211147 | 1.4566 |
| 6 | 241.907 | 9380 | 23807 | 1 | 1 | 0.0432836 | 0.047339 | 4.5472 |
| 7 | 229.321 | 8558 | 22035 | 1 | 1 | 0.049778 | 0.0581348 | 2.1804 |
| 8 | 191.965 | 7943 | 21525 | 1 | 1 | 0.0168702 | 0.0244832 | 1.5628 |
| 9 | 238.108 | 9212 | 24077 | 1 | 1 | 0.0443986 | 0.0463098 | 2.0999 |
| 10 | 242.936 | 9386 | 24420 | 1 | 1 | 0.0307905 | 0.033579 | 1.6465 |
| 11 | 95.3909 | 3354 | 9525 | 0.6102 | 0.5725 | 0.0405486 | 0.0350656 | 2.0966 |

Network v2 per lacuna

| lacuna_id | ring_attached_length_r30_px | ring_attached_length_r60_px | ring_length_w_r30_px | ring_length_w_r60_px | sholl_crossings_r10 | sholl_crossings_r20 | sholl_crossings_r30 |
|---|---|---|---|---|---|---|---|
| 1 | 307 | 905 | 385.433 | 1221.61 | 8 | 13 | 17 |
| 2 | 528 | 1231 | 659.156 | 1624.61 | 18 | 16 | 19 |
| 3 | 358 | 1014 | 424.149 | 1175.66 | 10 | 12 | 18 |
| 4 | 449 | 914 | 514.889 | 1109.35 | 14 | 14 | 9 |
| 5 | 136 | 225 | 169.125 | 491.546 | 4 | 6 | 8 |
| 6 | 394 | 972 | 452.877 | 1275.63 | 10 | 13 | 14 |
| 7 | 371 | 1180 | 490.617 | 1457.24 | 11 | 15 | 20 |
| 8 | 96 | 272 | 147.569 | 589.446 | 3 | 5 | 7 |
| 9 | 376 | 979 | 462.191 | 1254.86 | 12 | 14 | 16 |
| 10 | 222 | 600 | 331.321 | 931.98 | 8 | 12 | 13 |
| 11 | 119 | 302 | 152.225 | 375.664 | 4 | 4 | 6 |


543_3: distributions, pre-validation, px
Lacuna area (n = 11)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 38294)
count
width (px)